

Report No. ATSU240601911



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检测
TESTING
CNAS L9762



UN38.3 测试报告

UN38.3 Test Report

产品名称	: UPS不间断电源
Product name	: UNINTERRUPTIBLE POWER SUPPLY (UPS)
型号规格	: UPS 1000VA Pro/800W
Model/Type	: Battery specification: 25.6V, 9Ah, 230.4Wh
委托方	: 山东精工电源科技有限公司
Client	: Shandong Goldencell Power Technology Co., Ltd.

东莞市全测电子科技有限公司
ATS Electronic Technology Co., Ltd.

测试总览 Test Summary	
产品名称 Product name	UPS不间断电源 UNINTERRUPTIBLE POWER SUPPLY (UPS)
型号规格 Model/Type	UPS 1000VA Pro/800W Battery specification: 25.6V, 9Ah, 230.4Wh
商标 Trade mark	不适用 N/A
锂含量 Lithium Content	/
样品外观颜色 Appearance	黑色+白色 Black+ White
委托方 Client	山东精工电源科技有限公司 Shandong Goldencell Power Technology Co., Ltd.
委托方地址 Client Address	山东省枣庄高新区光明路X6699号 No.X6699 Guangming Boulevard, Hightech District, Zaozhuang City, Shandong Province, China
生产厂家 Manufacturer	山东精工电源科技有限公司 Shandong Goldencell Power Technology Co., Ltd.
生产厂家地址 Manufacturer Address	山东省枣庄高新区光明路X6699号 No.X6699 Guangming Boulevard, Hightech District, Zaozhuang City, Shandong Province, China
样品数量 Quantity of sample	电池: 16pcs 电芯: 30pcs
样品编号 Sample No.	ATSP2305066A A-001~A-016 ATSP2305066A A-017~A-046
测试标准 Testing standard	联合国《试验和标准手册》第七修订版修正1, 第38.3节: 锂金属和锂离子电池组 UNITED NATIONS "Manual of Tests and Criteria", Seventh revised edition Amendment 1(ST/SG/AC.10/11/Rev.7/Amend.1), Section 38.3: Lithium metal and lithium ion batteries
接样日期 Received date	2024-05-30
测试周期 Test period	2023-06-30 to 2023-07-14
备注 Remark: 此报告（报告编号ATSU240601911）只是在报告（报告编号ATSU230506611）的基础上修改了型号和标签，其他均保持一致。 This report (Report No. ATSU240601911) only modifies the model and label on the basis of the report (report No. ATSU230506611), and everything else remains the same.	

测试结论 Test Conclusion					
章节 Clause	测试项目名称 Name of test	样品编号 Sample No.	样品状态 Sample Condition	结论 Conclusion	备注 Remarks
38.3.4.1	试验T.1 高度模拟 Test T.1 Altitude simulation	ATSP2305066A A-001~A-004	第一个交替充电放电周期完全充电 First cycle in fully charged states	通过 Pass	--
		ATSP2305066A A-005~A-008	第二十五个交替充电放电周期完全充电 After 25 cycles ending in fully charged states		
38.3.4.2	试验T.2 温度试验 Test T.2 Thermal test	ATSP2305066A A-001~A-004	第一个交替充电放电周期完全充电 First cycle in fully charged states	通过 Pass	--
		ATSP2305066A A-005~A-008	第二十五个交替充电放电周期完全充电 After 25 cycles ending in fully charged states		
38.3.4.3	试验T.3 振动 Test T.3 Vibration	ATSP2305066A A-001~A-004	第一个交替充电放电周期完全充电 First cycle in fully charged states	通过 Pass	--
		ATSP2305066A A-005~A-008	第二十五个交替充电放电周期完全充电 After 25 cycles ending in fully charged states		
38.3.4.4	试验T.4 冲击 Test T.4 Shock	ATSP2305066A A-001~A-004	第一个交替充电放电周期完全充电 First cycle in fully charged states	通过 Pass	--
		ATSP2305066A A-005~A-008	第二十五个交替充电放电周期完全充电 After 25 cycles ending in fully charged states		
38.3.4.5	试验T.5 外部短路 Test T.5 External Short circuit	ATSP2305066A A-001~A-004	第一个交替充电放电周期完全充电 First cycle in fully charged states	通过 Pass	--
		ATSP2305066A A-005~A-008	第二十五个交替充电放电周期完全充电 After 25 cycles ending in fully charged states		
38.3.4.6	试验T.6 撞击/挤压 Test T.6 Impact/Crush	ATSP2305066A A-017~A-021	第一个交替充电放电周期充电至标称容量的50%状态 At first cycle at 50% of the design rated capacity	通过 Pass	圆柱形电芯直径为26.34mm / Cylindrical cell is 26.34mm in diameter.
		ATSP2305066A A-022~A-026	第二十五个交替充电放电周期充电至标称容量的50%状态 After 25 cycles ending at 50% of the design rated capacity		

38.3.4.7	试验T.7 过度充电 Test T.7 Overcharge	ATSP2305066A A-009~A-012	第一个交替充电放电周期完全充电 First cycle in fully charged states	通过 Pass	--
		ATSP2305066A A-013~A-016	第二十五个交替充电放电周期完全充电 After 25 cycles ending in fully charged states		
38.3.4.8	试验T.8 强制放电 Test T.8 Forced discharge	ATSP2305066A A-027~A-036	第一个交替充电放电周期完全放电 First cycle in fully discharged states	通过 Pass	--
		ATSP2305066A A-037~A-046	第二十五个交替充电放电周期完全放电 After 25 cycles, ending in fully discharged states		

检验结论/ Test Conclusion:

由山东精工电源科技有限公司送检的UPS不间断电源, 型号UPS 1000VA Pro/800W, 依据联合国《试验和标准手册》第七修订版修正1, 第38.3节: 锂金属和锂离子电池组进行全项目测试。当采用准确度方法判定规则时, 被测样品符合规范的要求。

The UNINTERRUPTIBLE POWER SUPPLY (UPS), Model UPS 1000VA Pro/800W, submitted by Shandong Goldencell Power Technology Co., Ltd. is tested according to UNITED NATIONS "Manual of Tests and Criteria", Seventh revised edition Amendment 1(ST/SG/AC.10/11/Rev.7/Amend.1), Section 38.3: Lithium metal and lithium ion batteries. The test items are full items. The sample received complies with Specification when Accuracy Method decision rule is applied.

测试结果: 通过。

The test results: Pass.

签发日期: 2024-06-05
Date of issue:

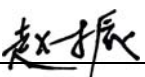
检测:

Tested by:

签字:

Signature:

赵振



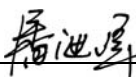
审核:

Reviewed by:

签字:

Signature:

潘海峰



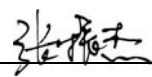
批准:

Approved by:

签字:

Signature:

张振杰





T.1 高度模拟 Altitude simulation							
测试方法/ Test Method							
将测试样品放在温度为$20\pm5^{\circ}\text{C}$，大气压力为不大于11.6kPa的环境中贮存不少于6个小时。对样品在测试前后进行称重，并记录电压。 The samples were stored for at least 6 hours at a pressure of 11.6kPa or less and a temperature of $20\pm5^{\circ}\text{C}$. The samples were weighed before and after the exposure. The samples voltage was also determined before and after the test.							
测试结果/ Test Results							
样品编号 Sample No.	测试前质量 (克) Weight Before Test (g)	测试后质量 (克) Weight After Test (g)	质量损失% Percentage of Weight Loss	测试前电压 (伏) Voltage Before Test (V)	测试后电压 (伏) Voltage After Test (V)	残余电压% Percentage of Residual Voltage	结果 Results
ATSP2305066A A-001	5667.55	5666.25	0.023	26.56	26.55	99.96	(6) (7)
ATSP2305066A A-002	5666.30	5665.50	0.014	26.55	26.54	99.96	(6) (7)
ATSP2305066A A-003	5667.15	5666.10	0.019	26.53	26.52	99.96	(6) (7)
ATSP2305066A A-004	5668.65	5668.10	0.010	26.55	26.54	99.96	(6) (7)
ATSP2305066A A-005	5667.70	5666.85	0.015	26.54	26.53	99.96	(6) (7)
ATSP2305066A A-006	5666.40	5665.60	0.014	26.56	26.55	99.96	(6) (7)
ATSP2305066A A-007	5668.35	5667.40	0.017	26.53	26.52	99.96	(6) (7)
ATSP2305066A A-008	5668.20	5667.10	0.019	26.55	26.54	99.96	(6) (7)
备注/Remark: 产品只有AC输出和AC输入，测试前电压和测试后电压需要从内部电池组进行测量。 The product has AC output and AC input only. The voltage before test and after test need to be measured from the internal battery pack. 结果/ Result: (1) 漏液/ Leakage (2) 排气/ Venting (3) 解体/ Disassembly (4) 破裂/ Rupture (5) 起火/ Fire (6) 无漏液、无排气、无解体、无破裂、无起火/ No leakage, no venting, no disassembly, no rupture and no fire (7) 开路电压不低于试验前开路电压的90%/ The open circuit voltage of each cell after testing was greater than 90%							

T.2 温度试验 Thermal test							
测试方法/ Test Method							
测试样品将进行如下温度循环测试。样品测试前后进行称重，并记录电压。 The samples were subjected to temperature cycling consisting of the following. The samples were weighed before and after the exposure. The samples voltage was also determined before and after the test.							
样品进箱 Samples In	烤箱温度在30分钟内上升到72±2°C，并维持此温度X*小时 The chamber temperature was raised to 72±2°C within 30 minutes and maintained at this temperature for X* hours.						
	烤箱温度在30分钟内降低到-40±2°C，并维持此温度X*小时 The chamber temperature was reduced to -40±2°C within 30 minutes and maintained at this temperature for X* hours.						
	重复此顺序测试额外9个循环（总共10个循环） Repeat the sequence for 9 additional cycles (total of 10 cycles).						
样品出箱 Samples Out	在第10个循环后，于20±5°C环境下储存24小时，然后检查其状态 After the 10th cycle, store the batteries at ambient temperature 20±5°Cfor 24 hours prior to examination.						
注：样品承受极端温度的持续时间（X*）按如下确定： Note: The duration of exposure to the test temperature extremes(X*) was determined as below:							
[] 小电芯和小电池为6小时/ Small cells and small batteries: 6 hours							
[X] 大电芯和大电池为12小时/ Large cells and large batteries: 12 hours							
测试结果/ Test Results							
样品编号 Sample No.	测试前质量（克） Weight Before Test (g)	测试后质量（克） Weight After Test (g)	质量损失% Percentage of Weight Loss	测试前电压（伏） Voltage Before Test (V)	测试后电压（伏） Voltage After Test (V)	残余电压% Percentage of residual Voltage	结果 Results
ATSP2305066A A-001	5666.25	5664.90	0.024	26.55	26.30	99.06	(6) (7)
ATSP2305066A A-002	5665.50	5664.05	0.026	26.54	26.31	99.13	(6) (7)
ATSP2305066A A-003	5666.10	5664.60	0.026	26.52	26.28	99.10	(6) (7)
ATSP2305066A A-004	5668.10	5666.35	0.031	26.54	26.29	99.06	(6) (7)
ATSP2305066A A-005	5666.85	5665.30	0.027	26.53	26.29	99.10	(6) (7)
ATSP2305066A A-006	5665.60	5664.55	0.019	26.55	26.30	99.06	(6) (7)
ATSP2305066A A-007	5667.40	5666.40	0.018	26.52	26.30	99.17	(6) (7)
ATSP2305066A A-008	5667.10	5665.70	0.025	26.54	26.31	99.13	(6) (7)
备注/Remark: 产品只有AC输出和AC输入，测试前电压和测试后电压需要从内部电池组进行测量。 The product has AC output and AC input only. The voltage before test and after test need to be measured from the internal battery pack. 结果/ Result: (1) 漏液/ Leakage (2) 排气/ Venting (3) 解体/ Disassembly (4) 破裂/ Rupture (5) 起火/ Fire (6) 无漏液、无排气、无解体、无破裂、无起火/ No leakage, no venting, no disassembly, no rupture and no fire (7) 开路电压不低于试验前开路电压的90%/ The open circuit voltage of each cell after testing was greater than 90%							

T.3 振动 Vibration							
测试方法/ Test Method							
测试样品将进行如下振动测试。样品测试前后进行称重，并记录电压。 The samples were subjected to vibration tests consisting of the following. The samples were weighed before and after the exposure. The samples voltage was also determined before and after the test. 测试样品牢固地安装在振动台上。振动以正弦波形式，以7Hz增加至200Hz，然后在减少回到7Hz为一个循环，一个循环持续15分钟的对数前移传送。以振动的其中一个方向必须是垂直样品极性，对每个样品从三个互相垂直的方向上循环12次，每个方向3个小时。 The samples were firmly secured to the platform of the vibration machine without distorting the sample in such a manner as to faithfully transmit the vibration. The vibration was a sinusoidal waveform with a logarithmic sweep between 7 Hz and 200 Hz and back to 7 Hz traversed in 15 minutes. This cycle was repeated 12 times for a total of 3 hours for each of three mutually perpendicular mounting positions of the sample. One of the directions of vibration was perpendicular to the terminal face. 对数扫频如下/ The logarithmic frequency sweep was as follows: [] 对于电芯和小电池：7赫兹开始保持1gn的最大加速度直到频率为18赫兹，然后将振幅保持在0.8毫米（总偏移1.6毫米）并增加频率直到最大加速度达到8gn（频率约为50赫兹），将最大加速度保持在8gn直到频率增加到200赫兹。 For cells and small batteries: From 7Hz a peak acceleration of 1gn was maintained until 18Hz is reached. The amplitude was then maintained at 0.8mm (1.6mm total excursion) and the frequency increased until a peak acceleration of 8gn occurred (approximately 50Hz). A peak acceleration of 8gn was then maintained until the frequency was increase to 200Hz. [X] 对大电池：7赫兹开始保持1gn的最大加速度直到频率为18赫兹，然后将振幅保持在0.8毫米（总偏移1.6毫米）并增加频率直到最大加速度达到2gn（频率约为25赫兹），将最大加速度保持在2gn直到频率增加到200赫兹。 For large batteries: From 7Hz a peak acceleration of 1gn was maintained until 18Hz is reached. The amplitude was then maintained at 0.8mm (1.6mm total excursion) and the frequency increased until a peak acceleration of 2gn occurred (approximately 25Hz). A peak acceleration of 2gn was then maintained until the frequency was increase to 200Hz.							
测试结果/ Test Results							
样品编号 Sample No.	测试前质量（克） Weight Before Test (g)	测试后质量（克） Weight After Test (g)	质量损失% Percentage of Weight Loss	测试前电压（伏） Voltage Before Test (V)	测试后电压（伏） Voltage After Test (V)	残余电压% Percentage of residual Voltage	结果 Results
ATSP2305066A A-001	5664.90	5664.40	0.009	26.30	26.30	100.00	(6) (7)
ATSP2305066A A-002	5664.05	5663.45	0.011	26.31	26.31	100.00	(6) (7)
ATSP2305066A A-003	5664.60	5664.10	0.009	26.28	26.28	100.00	(6) (7)
ATSP2305066A A-004	5666.35	5665.15	0.021	26.29	26.29	100.00	(6) (7)
ATSP2305066A A-005	5665.30	5664.20	0.019	26.29	26.29	100.00	(6) (7)
ATSP2305066A A-006	5664.55	5663.95	0.011	26.30	26.29	99.96	(6) (7)
ATSP2305066A A-007	5666.40	5665.35	0.019	26.30	26.29	99.96	(6) (7)
ATSP2305066A A-008	5665.70	5664.35	0.024	26.31	26.31	100.00	(6) (7)
备注/Remark: 产品只有AC输出和AC输入，测试前电压和测试后电压需要从内部电池组进行测量。 The product has AC output and AC input only. The voltage before test and after test need to be measured from the internal battery pack. 结果/ Result: (1) 漏液/ Leakage							

- (2) 排气/ Venting
- (3) 解体/ Disassembly
- (4) 破裂/ Rupture
- (5) 起火/ Fire
- (6) 无漏液、无排气、无解体、无破裂、无起火/ No leakage, no venting, no disassembly, no rupture and no fire
- (7) 开路电压不低于试验前开路电压的90%/ The open circuit voltage of each cell after testing was greater than 90%

T.4 冲击 Shock

测试方法/ Test Method

样品将进行如下冲击测试。对样品在测试前后进行称重，并记录电压。以稳固的托架固定住每个电芯和电池样品全部配件表面。每个样品将进行如下半正弦冲击测试：

The samples were subjected to shock. The samples were weighed before and after the exposure. The samples voltage was also determined before and after the test. The sample was secured to the testing machine by means of a rigid mount, which supports all mounting surfaces of the sample. Each sample was subjected to a half-sine shock as below:

[] 小电芯：峰值为150gn，脉冲持续6毫秒。

For cells: Peak acceleration of 150gn and pulse duration of 6 milliseconds.

[] 大电芯：峰值为50gn，脉冲持续11毫秒。

For large cells: Peak acceleration of 50gn and pulse duration of 11 milliseconds.

[] 小电池：取如下较小值为峰值，脉冲持续6毫秒。

For small batteries: Peak acceleration of the smaller of the following, and pulse duration of 6 milliseconds.

Battery	Minimum peak acceleration	Pulse duration
	150gn or result of formula	
Small batteries	$\text{Acceleration (gn)} = \sqrt{\left(\frac{100850}{\text{mass}^*}\right)}$ Whichever is smaller	6ms

[X] 大电池：取如下较小值为峰值，脉冲持续11毫秒。

For large batteries: Peak acceleration of the smaller of the following, and pulse duration of 11 milliseconds.

Battery	Minimum peak acceleration	Pulse duration
	50gn or result of formula	
Large batteries	$\text{Acceleration (gn)} = \sqrt{\left(\frac{30000}{\text{mass}^*}\right)}$ Whichever is smaller	11ms

每个测试样品须在三个互相垂直的电池安装方位的正方向经受三次冲击，接着在反方向经受三次冲击，总共经受18次冲击。

Each sample was subjected to three shocks in the positive direction followed by three shocks in the negative direction of three mutually perpendicular mounting positions of the cell for a total of 18 shocks.

测试结果/ Test Results

样品编号 Sample No.	测试前质量 (克) Weight Before Test (g)	测试后质量 (克) Weight After Test (g)	质量损失% Percentage of Weight Loss	测试前电压 (伏) Voltage Before Test (V)	测试后电压 (伏) Voltage After Test (V)	残余电压% Percentage of residual Voltage	结果 Results
ATSP2305066A A-001	5664.40	5664.40	0.000	26.30	26.30	100.00	(6) (7)
ATSP2305066A A-002	5663.45	5663.45	0.000	26.31	26.31	100.00	(6) (7)
ATSP2305066A A-003	5664.10	5664.10	0.000	26.28	26.28	100.00	(6) (7)
ATSP2305066A A-004	5665.15	5665.15	0.000	26.29	26.29	100.00	(6) (7)
ATSP2305066A A-005	5664.20	5664.20	0.000	26.29	26.29	100.00	(6) (7)
ATSP2305066A A-006	5663.95	5663.30	0.011	26.29	26.29	100.00	(6) (7)
ATSP2305066A A-007	5665.35	5664.80	0.010	26.29	26.29	100.00	(6) (7)
ATSP2305066A A-008	5664.35	5663.65	0.012	26.31	26.31	100.00	(6) (7)

备注/Remark:

产品只有AC输出和AC输入，测试前电压和测试后电压需要从内部电池组进行测量。

The product has AC output and AC input only. The voltage before test and after test need to be measured from the internal battery pack.

结果/ Result:

- (1) 漏液/ Leakage
- (2) 排气/ Venting
- (3) 解体/ Disassembly
- (4) 破裂/ Rupture
- (5) 起火/ Fire
- (6) 无漏液、无排气、无解体、无破裂、无起火/ No leakage, no venting, no disassembly, no rupture and no fire
- (7) 开路电压不低于试验前开路电压的90%/ The open circuit voltage of each cell after testing was greater than 90%

T.5 外部短路 External short circuit			
测试方法/ Test Method			
为使样品达到均匀稳定的初始温度：57±4°C，样品需在此环境下暴露一段时间。 The samples shall be heated for a period of time noted below, to reach a homogeneous stabilized temperature of 57±4°C, measured on the external case: [] 小电芯和小电池至少暴露6小时。Small cells and small batteries: 6 hours. [X] 大电芯和大电池至少暴露12小时。Large cells and large batteries: 12 hours. 然后将样品正负极用小于0.1欧姆的总电阻回路进行短路，直到： The samples were then subjected to a short circuit condition with a total external resistance of less than 0.1 ohm, until: [] 对于电芯和小电池：样品外表温度恢复到57±4°C之后保持短路状态1小时以上。 For cells and small batteries: 1 hour after the external case temperature of sample has returned to 57±4°C. [X] 对于大电池：样品表面温度下降所测最大温升的一半，并保持低于该数值。 For large batteries: After the external case temperature of sample has decreased by half of the maximum temperature increase observed during the test and remains below that value.			
测试结果/ Test Results			
样品编号 Sample No.	测试前电压（伏） Voltage Before Test(V)	最高温度(°C) Maximum Temperature, °C	结果 Results
ATSP2305066A A-001	26.30	57.8	(4) (5)
ATSP2305066A A-002	26.31	57.6	(4) (5)
ATSP2305066A A-003	26.28	57.5	(4) (5)
ATSP2305066A A-004	26.29	57.7	(4) (5)
ATSP2305066A A-005	26.29	57.8	(4) (5)
ATSP2305066A A-006	26.29	57.7	(4) (5)
ATSP2305066A A-007	26.29	57.6	(4) (5)
ATSP2305066A A-008	26.31	57.5	(4) (5)
备注/Remark: 产品只有AC输出和AC输入，测试前电压和测试后电压需要从内部电池组进行测量，且外部短路测试也是针对内部电池组进行的。 The product has AC output and AC input only. The voltage before test and after test need to be measured from the internal battery pack, and the external short-circuit test is also carried out for the internal battery pack. 结果/ Result: (1) 解体/ Disassembly (2) 破裂/ Rupture (3) 起火/ Fire (4) 测试后6小时内无解体，无破裂，无起火/ No disassembly, no rupture, no fire within 6 hours after the test (5) 最高温度不超过170摄氏度/ The maximum temperature did not exceed 170°C			

T.6 撞击/挤压 Impact/Crush

测试方法/ Test Method

- [X] 撞击（适用于直径不小于18毫米的圆柱形电芯）
Impact (for cylindrical cells greater not less than 18mm in diameter)
将试验样品放在一个平坦光滑的平面上。将一条316型不锈钢棒，其直径为 $15.8\text{mm} \pm 0.1\text{mm}$ ，长度为至少6cm，或电芯的最长边长度（两者中较大者），放置在样品中心。将一质量为 $9.1\text{kg} \pm 0.1\text{kg}$ 的物体于 $61\text{cm} \pm 2.5\text{cm}$ 的高度，无摩擦地从垂直滑轨落向样品。垂直滑轨与横向支承面互相垂直，保持90度。
A test sample was placed on a flat surface. A $15.8\text{mm} \pm 0.1\text{mm}$ diameter, at least 6cm long, or the longest dimension of the cell, whichever is greater, Type 316 stainless steel bar was placed across the center of the sample. A $9.1\text{kg} \pm 0.1\text{kg}$ mass was dropped from a height of $61\text{cm} \pm 2.5\text{cm}$ at the intersection of the bar and sample in a controlled manner, using a near frictionless, vertical sliding track or channel with minimal drag on the falling mass. The vertical track or channel used to guide the falling mass was oriented 90 degrees from the horizontal supporting surface.
接受撞击的试样，纵轴应与平坦的表面平行并与横放在试样中心的直径 $15.8\text{mm} \pm 0.1\text{mm}$ 弯曲表面的纵轴垂直。每一个试样只经受一次撞击。
The test sample was impacted with its longitudinal axis parallel to the flat surface and perpendicular to the longitudinal axis of a $15.8\text{mm} \pm 0.1\text{mm}$ diameter curved surface lying across the center of the test sample. Separate samples were used for each test.
- [] 挤压（适用于棱柱形、袋装、硬币/纽扣电池和直径小于18毫米的圆柱形电芯）
Crush (for prismatic, pouch, coin/button cells and cylindrical cells less than 18mm in diameter)
将样品放在两个平面之间挤压。挤压力度逐渐加大，在第一个接触点上的速度大约为1.5厘米/秒。挤压持续进行，直到出现以下三种情况之一：
A sample was crushed between two flat surfaces. The crushing was gradual with a speed of approximately 1.5 cm/s at the first point of contact. The crushing was continued until the first of the three options below has reached:
 - 施加的力达到 $13\text{kN} \pm 0.78\text{kN}$;
The applied force reaches $13\text{kN} \pm 0.78\text{kN}$
 - 电池的电压下降至少100毫伏，或者
The voltage of the cell drops by at least 100mV; or
 - 电池变形达到原始厚度的50%以上
The cell is deformed by 50% or more of its original thickness.
棱柱形或袋装电池应从最宽的一面施压。纽扣/硬币形电池应从其平坦表面施压。圆柱形应从与纵轴垂直的方向施压。
A prismatic or pouch cell was crushed by applying the force to the widest side. A button/coin cell was crushed by applying the force on its flat surfaces. For cylindrical cells, the crush force was applied perpendicular to the longitudinal axis
测试样品进一步观察6小时。未进行过其他测试的样品用于此测试。
The test sample was observed for a further 6 hours. Separate samples that have not previously been subjected to other tests were used for each test

测试结果/ Test Results

样品编号 Sample No.	测试前电压（伏） Voltage Before Test (V)	最高温度(°C) Maximum Temperature, °C	结果 Results
ATSP2305066A A-017	3.280	23.7	(3) (4)
ATSP2305066A A-018	3.281	24.1	(3) (4)
ATSP2305066A A-019	3.303	23.9	(3) (4)
ATSP2305066A A-020	3.310	23.9	(3) (4)
ATSP2305066A A-021	3.276	23.7	(3) (4)
ATSP2305066A A-022	3.275	23.8	(3) (4)

ATSP2305066A A-023	3.284	24.1	(3) (4)
ATSP2305066A A-024	3.273	23.9	(3) (4)
ATSP2305066A A-025	3.310	23.6	(3) (4)
ATSP2305066A A-026	3.303	23.8	(3) (4)
结果/ Result: (1) 解体/ Disassembly (2) 起火/ Fire (3) 测试后6小时内无解体, 无起火/ No disassembly, no fire within 6 hours after the test (4) 最高温度不超过170摄氏度/ The maximum temperature did not exceed 170°C			

T.7 过度充电 Overcharge			
测试方法/ Test Method			
2倍制造厂推荐的最大持续充电电流对样品充电。 Batteries were subjected to a charge current of twice the manufacturer's recommended maximum continuous charge current. 最小的测试电压由按如下决定/ The minimum voltage of the test was as follows: [] 如果厂家推荐的充电电压不超过18V, 本测试的最小充电电压应是厂家标定最大充电电压的两倍或者是22V之中的较小者。 When the manufacturer's recommended charge voltage is not more than 18V, the minimum voltage of the test was the lesser of 2 times the maximum charge voltage of the battery or 22V. [X] 如果厂家推荐的充电电压超过18V, 本测试的最小充电电压应是厂家标定的最大充电电压的1.2倍 When the manufacturer's recommended charge voltage is more than 18V, the minimum voltage of the test was 1.2 times the maximum charge voltage. 测试在20±5°C的环境温度下进行, 试验持续24小时。 Tests were conducted at ambient temperature 20±5°C. The duration of the test was 24 hours. 过充电流/ Overcharge Current 18000mA 过充电压/ Overcharge Voltage 34.56V			
测试结果/ Test Results			
样品编号 Sample No.	测试前电压 (伏) Voltage Before Test(V)	测量的过充电流 (毫安) Measured Overcharge Current (mA)	结果 Results
ATSP2305066A A-009	26.55	18000	(3)
ATSP2305066A A-010	26.53	18000	(3)
ATSP2305066A A-011	26.54	18000	(3)
ATSP2305066A A-012	26.56	18000	(3)
ATSP2305066A A-013	26.54	18000	(3)
ATSP2305066A A-014	26.54	18000	(3)
ATSP2305066A A-015	26.54	18000	(3)
ATSP2305066A A-016	26.55	18000	(3)
备注/Remark: 产品只有AC输出和AC输入, 测试前电压和测试后电压需要从内部电池组进行测量, 且外部短路测试也是针对内部电池组进行的。 The product has AC output and AC input only. The voltage before test and after test need to be measured from the internal battery pack, and the external short-circuit test is also carried out for the internal battery pack. 结果/ Result: (1) 解体/ Disassembly (2) 起火/ Fire (3) 测试后7天内无解体, 无起火/ No disassembly, no fire within seven days after the test			

T.8 强制放电 Forced discharge			
测试方法/ Test Method			
在常温环境下，将单个电芯连接在12V的直流电源上进行强制放电，此直流电源提供给每个电芯初始电流为制造厂指定的最大放电电流。 Each cell was forced discharged at ambient temperature by connecting it in series with a 12V DC power supply at an initial current equal to the maximum discharge current specified by the manufacturer. 指定的放电电流通过串联在测试电芯上的合适大小和功率的负载来获得，每个电芯的强制放电时间（小时）为额定容量除以初始电流（安培） The specified discharge current was obtained by connecting a resistive load of the appropriate size and rating in series with the test cell. Each cell was forced discharged for a time interval (in hours) equal to its rated capacity divided by the initial test current (in amperes).			
测试结果/ Test Results			
样品编号 Sample No.	测试前电压（伏） Voltage Before Test (V)	测试后电压（伏） Voltage After Test (V)	结果 Results
ATSP2305066A A-027	2.769	0.073	(3)
ATSP2305066A A-028	2.775	0.049	(3)
ATSP2305066A A-029	2.764	0.020	(3)
ATSP2305066A A-030	2.784	0.091	(3)
ATSP2305066A A-031	2.772	0.013	(3)
ATSP2305066A A-032	2.759	0.008	(3)
ATSP2305066A A-033	2.761	0.050	(3)
ATSP2305066A A-034	2.765	0.037	(3)
ATSP2305066A A-035	2.777	0.043	(3)
ATSP2305066A A-036	2.771	0.022	(3)
ATSP2305066A A-037	2.756	0.019	(3)
ATSP2305066A A-038	2.744	0.085	(3)
ATSP2305066A A-039	2.766	0.076	(3)
ATSP2305066A A-040	2.747	0.023	(3)
ATSP2305066A A-041	2.755	0.009	(3)
ATSP2305066A A-042	2.762	0.135	(3)
ATSP2305066A A-043	2.782	0.048	(3)
ATSP2305066A A-044	2.778	0.026	(3)
ATSP2305066A A-045	2.782	0.023	(3)
ATSP2305066A A-046	2.754	0.010	(3)
结果/Result: (1) 解体/ Disassembly (2) 起火/ Fire (3) 测试后7天内无解体，无起火/ No disassembly, no fire within seven days after the test			

样品照片
Photos of Samples

UPS不间断电源 / UNINTERRUPTIBLE POWER SUPPLY (UPS) (25.6V, 9Ah, 230.4Wh)



Figure 1 Front view of battery



Figure 2 Back view of battery

样品照片
Photos of Samples

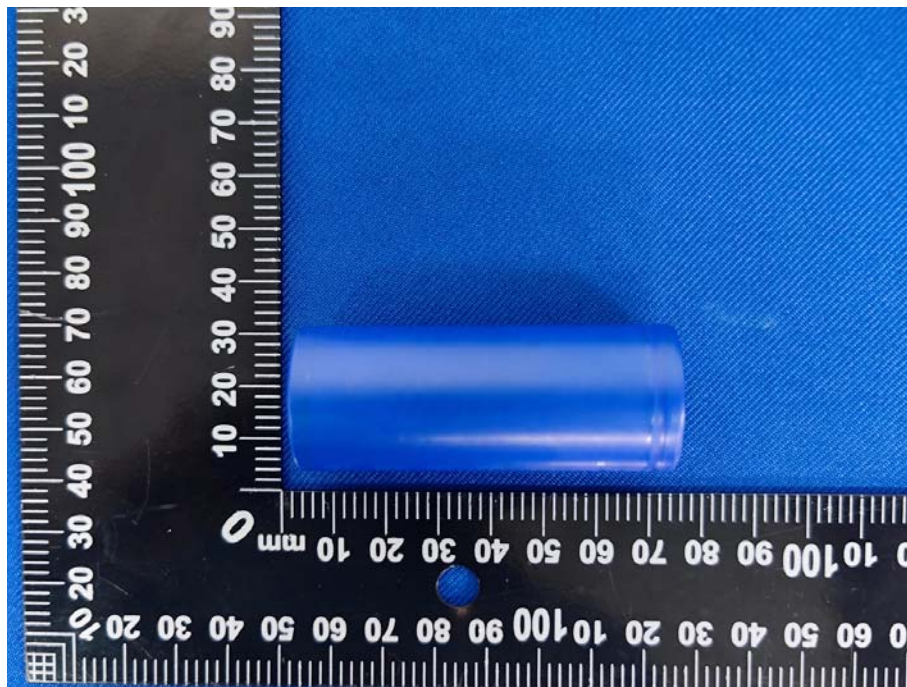


Figure 3 Side view of cell

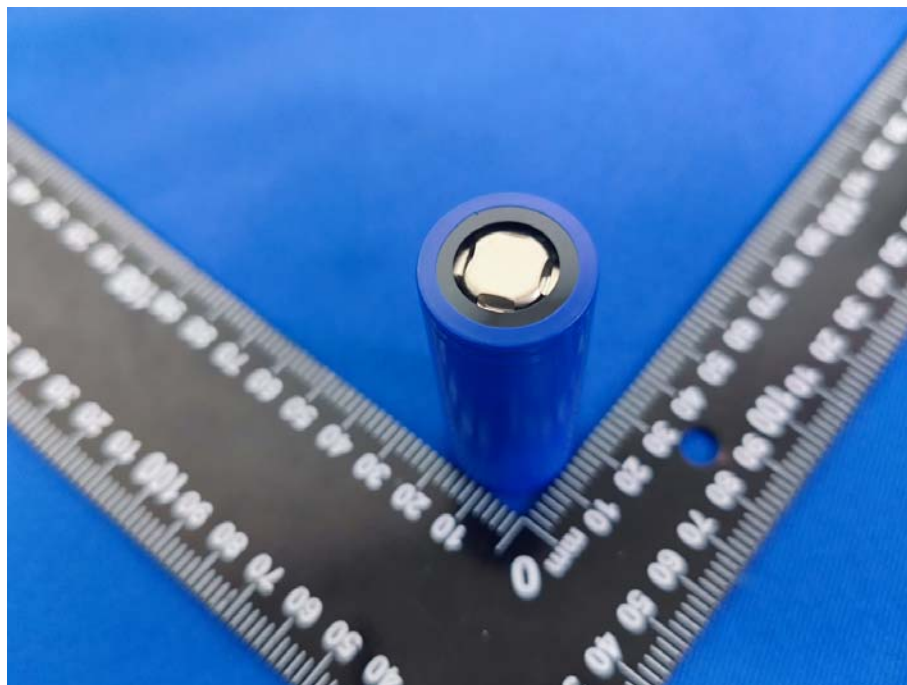


Figure 4 Top view of cell

样品照片
Photos of Samples

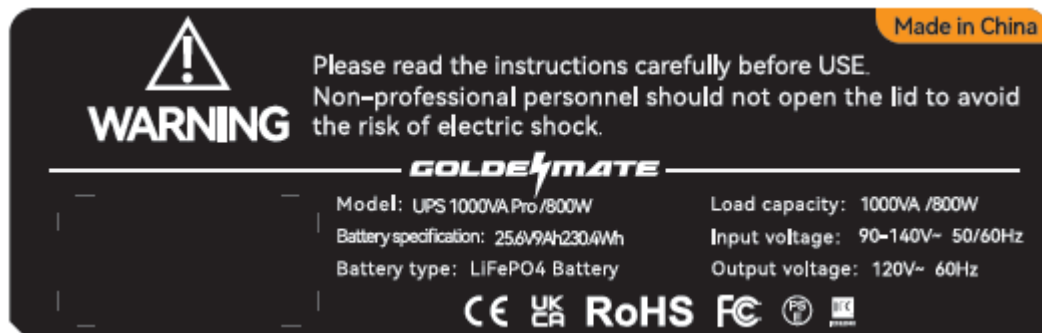


Figure 5 Battery label

注 意 事 项

Important

1. 未经本实验室书面同意，不得复制或部分地复制本报告。
Nobody is allowed to photocopy or partly photocopy this test report without written permission of ATS Electronic Technology Co., Ltd.
2. 本报告无批准人、审核人及检测人签名无效。
This test report is invalid without the signatures of Approver, Reviewer and Tester.
3. 本报告涂改无效。
The test report is invalid if altered.
4. 本报告结果仅对送检样品负责。
The test report is valid for the tested samples only.
5. 本报告中以点号代替小数点。
Throughout this report a point is used as the decimal separator.

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